A uniform point-mass bound for sums of prescribed-rank matrices

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Abstract

Independent uniform matrices of prescribed ranks over a finite field have a uniformly bounded density relative to the uniform measure as soon as the sum of the ranks is at least twice the matrix dimension. An explicit absolute bound proves Conjecture 5.5 of Gill, Lifshitz, Pyber and Szabó, with $K(\epsilon) = \lceil 2/\epsilon \rceil$. The proof uses rectangular factorization and a rank-sensitive estimate for the probability of full rank.

Task 1: the complete result

Let q be any prime power, $n \geq 1$, and let $R_{n,q}(r)$ denote the number of rank-r matrices in $\mathrm{Mat}_{n\times n}(\mathbb{F}_q)$. For a fixed matrix M of rank t, let $N_{n,q}(r_1, \ldots, r_k; t)$ count tuples $(X_1, \ldots, X_k)$ with $\operatorname{rank} X_i = r_i$ and $\sum_i X_i = M$. This number depends only on t, by invertible left and right multiplication.

Theorem 1. *For every prime power q, every $n \geq 1$, and every sequence of integers $0 \leq r_i \leq n$ such that $\sum_{i=1}^{k} r_i \geq 2n$, one has*

$$N_{n,q}(r_1, \ldots, r_k; t) \leq 32768\, q^{-n^2} \prod_{i=1}^{k} R_{n,q}(r_i) \qquad (0 \leq t \leq n).$$

Consequently Conjecture 5.5, equivalently Question 5.22, of [1] holds with $K(\epsilon) = \lceil 2/\epsilon \rceil$ and an absolute implied constant independent even of ϵ.

The conclusion is the upper bound requested in the source. It does not assert a two-sided estimate or convergence to the uniform distribution. For example, when $n = 1$, $q = 2$ and every $r_i = 1$, the sum is deterministic. This example is entirely consistent with the asserted upper bound.

Two elementary estimates

The full-rank and subspace-count formulas below are standard and are also recorded in Fact 5.23 of [1]. We include their estimates to make the argument self-contained.

Lemma 1. *If U is uniform in $\mathrm{Mat}_{n\times r}(\mathbb{F}_q)$, where $0 \leq r \leq n$, then*

$$p_n(r) := \mathbb{P}(\operatorname{rank} U = r) = \prod_{i=0}^{r-1} (1 - q^{i-n}) \geq 4^{-r/n}.$$

The same probability applies to full row rank in $\mathrm{Mat}_{r\times n}$.

Proof. Choose the columns successively. After i independent columns their span has q^i elements, proving the product formula. The numbers $\log(1-q^{i-n})$, $0 \le i < n$, are decreasing in i. Their first r terms therefore have average at least their overall average:

$$\frac{\log p_n(r)}{r} \ge \frac{\log p_n(n)}{n} \qquad (r \ge 1).$$

Moreover, using $\prod_j (1-a_j) \ge 1 - \sum_j a_j$ for $0 \le a_j \le 1$,

$$p_n(n) = \prod_{j=1}^{n} (1-q^{-j}) \ge \frac{1}{2} \prod_{j=2}^{n} (1-2^{-j}) \ge \frac{1}{4}.$$

The empty product covers $n = 1$ in the last estimate and $r = 0$ in the statement. Exponentiating proves the claim; transposition gives its row version. □

Lemma 2. *Let $R \ge 2n$. If $A \in \mathrm{Mat}_{n\times R}(\mathbb{F}_q)$ and $B \in \mathrm{Mat}_{R\times n}(\mathbb{F}_q)$ are independent and uniform, then*

$$\mathbb{P}(AB = M) \le 8q^{-n^2} \quad \textit{for every } M \in \mathrm{Mat}_{n\times n}(\mathbb{F}_q).$$

Proof. Conditional on A having rank $n-d$, the columns of AB are independent uniform elements of $\operatorname{im} A$. Thus their probability of equaling the prescribed columns of M is either zero or $q^{-n(n-d)}$.

There are at most $4q^{d(n-d)}$ subspaces of dimension d in $\mathbb{F}_q^n$. Indeed, count ordered independent d-tuples by at most q^{nd}, and divide by their number in $\mathbb{F}_q^d$, which is $q^{d^2} p_d(d) \ge q^{d^2}/4$ for $d \ge 1$. The same upper bound is valid for $d = 0$. For any fixed such subspace W, the probability that $W \subseteq \ker A^{\mathsf{T}}$ is q^{-dR}: each of the R independent columns must satisfy d independent linear equations. A union bound therefore gives

$$\mathbb{P}(\operatorname{rank} A = n-d) \le 4q^{-d(R-n+d)} \qquad (0 \le d \le n).$$

It follows that

$$\begin{aligned}
\mathbb{P}(AB = M) &\le \sum_{d=0}^{n} 4q^{-d(R-n+d)} q^{-n(n-d)} \\
&= 4q^{-n^2} \sum_{d=0}^{n} q^{-d(R-2n+d)} \\
&\le 4q^{-n^2} \sum_{d=0}^{\infty} 2^{-d^2} \le 8q^{-n^2},
\end{aligned}$$

where $R \ge 2n$, $q \ge 2$ and $d^2 \ge d$ were used. □

Proof of the uniform prescribed-rank bound

Proof of Theorem 1. Choose m minimally so that $R = \sum_{i=1}^{m} r_i \ge 2n$. Then $2n \le R < 3n$, since the preceding sum is less than $2n$ and $r_m \le n$. Independently choose uniform matrices

$$U_i \in \mathrm{Mat}_{n\times r_i}(\mathbb{F}_q), \qquad V_i \in \mathrm{Mat}_{r_i\times n}(\mathbb{F}_q), \qquad 1 \le i \le m,$$

and let E be the event that every U_i has full column rank and every V_i has full row rank. Lemma 1 gives

$$\mathbb{P}(E) = \prod_{i=1}^{m} p_n(r_i)^2 \ge 16^{-R/n} \ge 16^{-3}.$$

Conditional on E, the products U_iV_i are independent uniform rank-r_i matrices. To see uniformity, for any fixed rank-r matrix X, the columns of U must be an ordered basis of $\operatorname{im} X$. There are exactly $|\operatorname{GL}(r,q)|$ choices, and each determines a unique full-row-rank V with $UV = X$. The rank-zero case has one empty factorization. The conditioning event separates across the independent blocks, so independence also persists.

Before conditioning, form the matrices

$$A = [\,U_1 \ \cdots \ U_m\,], \qquad B = \begin{bmatrix} V_1 \\ \vdots \\ V_m \end{bmatrix}.$$

They are independent uniform matrices of the sizes in Lemma 2, and $AB = \sum_{i=1}^{m} U_iV_i$. Consequently, for independent uniform rank-r_i matrices X_i,

$$\mathbb{P}\left(\sum_{i=1}^{m} X_i = M\right) = \mathbb{P}(AB = M \mid E) \le \frac{\mathbb{P}(AB = M)}{\mathbb{P}(E)} \le 8 \cdot 16^3 q^{-n^2} = 32768 q^{-n^2}.$$

Adding the remaining independent matrices cannot increase the largest point probability: average this bound over their sum. Finally, multiply by $\prod_i R_{n,q}(r_i)$ to obtain the stated count.

If $r_i \ge \epsilon n$ and $k \ge \lceil 2/\epsilon \rceil$, then $\sum_i r_i \ge k\epsilon n \ge 2n$. This proves the conjecture for $0 < \epsilon \le 1$; for $\epsilon > 1$ its hypotheses are impossible. □

The rank-sensitive conditioning bound is what keeps the constant absolute. Using only $p_n(r) \ge 1/4$ separately for each block would introduce an unnecessary factor exponential in the number of summands. The source's established case of three ranks at least $2n/3$ is credited to its Theorem 5.3. The present argument proves the full many-rank extension and the stronger sum-of-ranks criterion.

Proposed follow-up Tasks 2–5

These are proposed tasks, not additional completed results.

2. Determine the smallest absolute constant valid under the sole condition $\sum_i r_i \ge 2n$.
3. Find sharp necessary rank-sum thresholds for uniform point-mass bounds, including sequences with $\sum_i r_i < 2n$.
4. Prove the corresponding optimal theorem for rectangular $a \times b$ matrices with arbitrary prescribed ranks.
5. Characterize the rank sequences and fields for which the normalized point probabilities converge uniformly to one, and obtain quantitative rates.

References

[1] N. Gill, N. Lifshitz, L. Pyber and E. Szabó, *Initiating the proof of the Liebeck–Nikolov–Shalev conjecture*, arXiv:2408.07800v2, 24 September 2024. Conjecture 5.5, Question 5.22, Theorem 5.3 and Fact 5.23.